

Final Examination
Academic Year 2025-2026

UID No:

Program Name/Code: Bachelor of Engineering- Chemical Engineering

Subject Code : 23CHT-305

Subject Title : Numerical methods

Semester :5

Time : 3 Hour

Maximum Marks : 60

Instructions :

- The question paper is consisting of three sections. It is compulsory for students to attempt all questions of Section A and Section B.
- Question no. 10 & 11 of section C, are compulsory to be attempted.
- Students to attempt any one question from question no. 12 & question no. 13 of section C

Q. No	Statement	CO Mapping	BT Level
Section A 5x2=10 marks			
1	If 0.333 is the approximate value of 1/3, find absolute and relative errors	CO1	2
2	Write Simpson 1/3 rule.	CO1	2
3	Write the normal equations of Fit a Parabola.	CO1	1
4	Explain Taylor expansion method for the numerical solution of ordinary differential equations	CO1	2
5	Explain modified Euler's method for the numerical solution of ordinary differential equations	CO1	1
Section B 4x5=20 marks			
6	Find by Newton's method, the root of the equation $3x = \cos x + 1$ correct to four decimal places.	CO2	3
7	Use Romberg's method to compute $\int_0^1 \frac{dx}{1+x^2}$ correct to four decimal places	CO2	4

8	Find by Taylor's series method, the values of y at x = 0.1 and x = 0.2 to five places of decimals from $\frac{dy}{dx} = x^2y - 1, y(0) = 1$	CO2	3
9	Using Eulers method find an approximate value of y corresponding to x=1, given that $\frac{dy}{dx} = x + y$ and y=1 when x= 0, taking h=0.1	CO2	4
Section C 3x10=30 marks			
10	Use Gauss's forward formula to evaluate y_{30} given that $y_{21} = 18.4780, y_{25} = 17.8144, y_{29} = 17.1070, y_{33} = 16.3432$ and $y_{37} = 15.5154$	CO3	5
11	Apply Gauss Jordan method to solve the equations $\begin{aligned} x + y + z &= 9 \\ 2x - 3y + 4z &= 13 \\ 3x + 4y + 5z &= 40 \end{aligned}$	CO3	5
Optional Question of Section C			
12	An experiment gave the following values: v: 350 400 500 600 t: 61 26 7 2.6 It is known that v and t are connected by the relation $v = at^b$. Find the best possible values of a and b.	CO3	4
OR			
13	Apply R – K of 4 th order method to find the solution of differentiate equation. $\frac{dy}{dx} = x + y^2$, at x = 15 with h =0.2, given that y = 1.5 & x = 1	CO3	4

CU UPDATES

Printed Pages:

UID No:

Final Examination
Academic year 2024–2025

Program Name/Code: BE AME

Subject Code: 22AET-358

Subject Title: Numerical Methods and Optimization Techniques

Semester: 6th

Time: 3 Hour

Maximum Marks: 60

Instructions:

- The question paper is consisting of three sections. It is compulsory for students to attempt all questions of Section A and Section B.
- Question no. 10 & 11 of section C, are compulsory to be attempted.
- Students to attempt any one question from question no. 12 & question no. 13 of section C

10	Using the bisection method, find an approximate root of the equation $\sin x = 1/x$, that lies between $x=1$ and $x=1.5$ (measured in radians). Carry out computations up to the 7th stage	CO2	BT5
11	Using modified Euler's method, find an approximate value of y when $x=0.3$, given that $dy/dx = x + y$ and $y=1$ when $x=0$.	CO3	BT5
Optional Question of Section C			
12	Apply the Runge-Kutta method to find the approximate value of y for $x=0.2$, in steps of 0.1, if $dy/dx = x + y^2$, $y=1$ where $x=0$.	CO4	BT5
OR			
13	Apply Milne's method, to find a solution of the differential equation $y' = x - y^2$ in the range $0 \leq x \leq 1$ for the boundary condition $y=0$ at $x=0$.	CO4	BT5

Q. No	Statement	CO Mapping	BT Level
Section A (5 x 2 = 10 marks)			
1	Explain the role of the function $f(x)$ in the false-position method?	CO5	BT2
2	Write the Picard iteration formula	CO3	BT1
3	Discuss the causes of round-off error in numerical computations.	CO4	BT2
4	Discuss the use of the average of forward and backward differences in central difference interpolation	CO3	BT1
5	Describe the approximation of the solution curve of an ODE using Euler's method	CO3	BT1
Section B (4 x 5 = 20 marks)			
6	Find a root of the equation $x^3 - 2x - 5 = 0$ using the secant method correct to three decimal places	CO4	BT3
7	Apply Bessel's formula to obtain y_{25} , given $y_{20}=2854$, $y_{24}=3162$, $y_{28}=3544$, $y_{32}=3992$.	CO5	BT4
8	Evaluate $\int_0^6 dx/1+x^2$ by Simpson's 1/3 rule.	CO3	BT4
9	Find the value of y for $x=0.1$ by Picard's method, given that $dy/dx = y - x/y + x$ when $y(0)=1$	CO2	BT4
Section C (3 x 10 = 30 marks)			

Final Examination

Academic Year 2024-2025

UID No:

Program Name/Code: BE MTC

Subject Code : 22MTT-323

Subject Title : Numerical Methods and Optimization Techniques

Semester : 6TH

Time : 3 Hour

Maximum Marks : 60

Instructions :

- The question paper is consisting of three sections. It is compulsory for students to attempt all questions of Section A and Section B.
- Question no. 10 & 11 of section C, are compulsory to be attempted.
- Students to attempt any one question from question no. 12 & question no. 13 of section C

Q. No	Statement	CO Mapping	BT Level
Section A 5x2=10 marks			
1	Find the relative error in taking $n = 3.141593$ as $22/7$.	CO2	2
2	When does the Gauss Elimination method fail to work?	CO2	2
3	What does the dual of a dual problem represent in linear programming?	CO2	1
4	If the primal problem's objective is to minimize, then what is the objective of dual problem?	CO2	2
5	How many allocations are required so that the solution of transportation problem with m rows (supplies) and n columns (destination) is feasible?	CO1	1
Section B 4x5=20 marks			
6	Evaluate $30^{-1/5}$ by Newton's iteration method, correct to four decimal places.	CO3	4
7	Solve $dy/dx = x + y$, $y(0) = 1$ by Taylor's series method. Hence find the values of y at $x = 0.1$.	CO4	3
8	Discuss the concept of the Initial Basic Feasible Solution (IBFS) in the Transportation Problem. Explain the North-West Corner Rule.	CO4	4
9		CO4	4

	Explain the Traveling Salesman Problem (TSP). What is the objective of the problem, and how does it differ from other optimization problems?																																												
	Section C 3x10=30 marks																																												
10	Using Newton divided difference formula, find the equation of the cubic curve which passes through the point (4, - 43), (7, 83), (9, 327), (12, 1053).	CO2	6																																										
11	Solve by Gauss-Seidel method the system of equations $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$.	CO4	5																																										
	Optional Question of Section C																																												
12	Use dual simplex method to solve the following LPP Minimize $z = 3x_1 + x_2$ Subject to $x_1 + x_2 \geq 1$, $2x_1 + 3x_2 \geq 2$; $x_1, x_2 \geq 0$.	CO5	6																																										
	OR																																												
13	ABC Ice Cream Company has a distribution depot in Greater Kailash Part I for distributing ice-cream in South Delhi. There are four vendors located in different parts of South Delhi (call them A, B, C and D) who have to be supplied ice-cream every day. The following matrix displays the distances (in kilometres) between the depot and the four vendors: <table> <tr> <td></td><td></td><td colspan="4">To</td></tr> <tr> <td></td><td>Depot</td><td>Vendor A</td><td>Vendor B</td><td>Vendor C</td><td>Vendor D</td></tr> <tr> <td>From Depot</td><td>–</td><td>3.5</td><td>3</td><td>4</td><td>2</td></tr> <tr> <td>Vendor A</td><td>3.5</td><td>–</td><td>4</td><td>2.5</td><td>3</td></tr> <tr> <td>Vendor B</td><td>3</td><td>4</td><td>–</td><td>4.5</td><td>3.5</td></tr> <tr> <td>Vendor C</td><td>4</td><td>2.5</td><td>4.5</td><td>–</td><td>4</td></tr> <tr> <td>Vendor D</td><td>2</td><td>3</td><td>3.5</td><td>4</td><td>–</td></tr> </table> What route should the company van follow so that the total distance travelled is minimized?			To					Depot	Vendor A	Vendor B	Vendor C	Vendor D	From Depot	–	3.5	3	4	2	Vendor A	3.5	–	4	2.5	3	Vendor B	3	4	–	4.5	3.5	Vendor C	4	2.5	4.5	–	4	Vendor D	2	3	3.5	4	–	CO5	3
		To																																											
	Depot	Vendor A	Vendor B	Vendor C	Vendor D																																								
From Depot	–	3.5	3	4	2																																								
Vendor A	3.5	–	4	2.5	3																																								
Vendor B	3	4	–	4.5	3.5																																								
Vendor C	4	2.5	4.5	–	4																																								
Vendor D	2	3	3.5	4	–																																								

CU UPDATES

Printed Pages: 2

UID No:

Final Examination
Academic year 2023–2024

Program Name/Code: BE-CSE/IT

Subject Code: 22CSH-259/22ITH-259

Subject Title: Numerical Methods and Optimization using Python

Semester: 4TH

Time: 3 Hour

Maximum Marks: 60

Instructions:

- The question paper is consisting of three sections. It is compulsory for students to attempt all questions of Section A and Section B.
- Question no. 10 & 11 of section C, are compulsory to be attempted.
- Students to attempt any one question from question no. 12 & question no. 13 of section C

Q. No	Statement	CO mapping
Section A (5 x 2 = 10 marks)		
1	Elaborate the fundamental principle of the Bisection method.	CO2
2	What is interpolation, and why is it used in data analysis?	CO4
3	Classify the optimization techniques.	CO3
4	What is the importance of handling both equality and inequality constraints in optimization problems?	CO4
5	Elucidate the fundamental principle of Simulated Annealing.	CO4
Section B (4 x 5 = 20 marks)		
6	For the given function $f(x) = x^3 - x - 1$, a real root lies in between the interval $[1, 2]$. Find the value of root till 5 iterations.	CO1
7	Find the unique polynomial $p(x)$ of degree 2 by using Lagrange's method of interpolation such that: $p(1)=1$, $p(3)=27$, $p(4)=64$	CO4
8	Describe the role of the fitness function in a Genetic algorithm. How the fitness function is defined, and how does it evaluate the quality of potential solutions?	CO3
9	Conduct four iterations of the secant method to locate a	CO3

	real root of the equation $f(x) = x^3 - 5x + 1 = 0$ within the specified interval $[0, 1]$.	
Section C (3 x 10 = 30 marks)		
10	Illustrate the fundamental principle for the Trapezoidal Rule with suitable example. Also discuss its appropriateness for accurate integration.	CO3
11	Solve the given LPP by graphical method and find the maxima for: $z=2x+5y$ The constraints are : $x+4y \leq 24$ $3x+y \leq 21$ $x+y \leq 9$ where $x, y \geq 0$	CO3
Optional Question of Section C		
12	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using (i) Simpson's $\frac{1}{3}$ rule taking $h = \frac{1}{4}$ (ii) Simpson's $\frac{3}{8}$ rule taking $h = \frac{1}{6}$	CO3
OR		
13	Apply Lagrange's Multiplier method to solve: $f(x,y)=xy$ with constraint $4x^2+y^2=8$	CO3

Printed Pages:

UID No:

End Semester Examination
Academic year Semester 2022–2023

Program Name/Code: Bachelor of Engineering- (Mechatronics)

Subject Code: 20MTT-377

Semester: 6th

Subject Title: Numerical Methods and Optimization Techniques

Time: 3 Hour

Maximum Marks: 60

	$2x+2y+7z = 19$	
12	Find y (32), using Newton's backward interpolation formulas if y (10) = 35.5, y (15) = 32.4, y (20) = 29.2, y (25) = 26.1, y (30) = 23.2, y (35) = 20.5.	CO5

Instructions: Attempt all questions

Q. No	Statement	CO mapping
	Section A 5 x 2 = 10 marks	
1	Write Bessel's formula.	CO5
2	Write Newton's Backward Interpolation Formula	CO5
3	Explain briefly Gauss-Jordan method.	CO4
4	Write Simpson 1/3 rule.	CO5
5	Round off to three significant figures: (i) 7.893 (ii) 12.865	CO1
	Section B 4 x 5 = 20 marks	
6	The rank of the matrix	CO4
7	Round off the following numbers correct to four significant figures: 3.26425, 35.46735, 4985561, 0.70035, 0.00032217	CO2
8	Show that the equations $5x+3y+7z=4$, $3x+26y+2z=9$, $7x+2y+10z=5$ are consistent and solve them by rank method.	CO4
9	Write a note on Newton-Raphson method.	CO2
	Section C 3 x 10 = 30 marks	
10	Find the root of the equation $x^3 - 4x - 9$ by using bisection method correct to three decimal places.	CO2
11	Solve the following system of equations by using Gauss-Seidel method $9x-2y+z = 50$ $X+5y-3z = 18$ -	CO4

Printed Pages:

UID No:

Final Examination
Academic year 2024–2025

Program Name/Code: Bachelor of Engineering- (Information Technology)

Subject Code: 21ITH-459/21CSH-459

Subject Title: Numerical Methods and Optimization using Python

Semester:7

Time: 3 Hour

Maximum Marks: 60

Instructions:

- The question paper is consisting of three sections. It is compulsory for students to attempt all questions of Section A and Section B.
- Question no. 10 & 11 of section C, are compulsory to be attempted.
- Students to attempt any one question from question no. 12 & question no. 13 of section C

Use of scientific calculator is allowed

Q. No	Statement	CO Mapping
Section A (5 x 2 = 10 marks)		
1	Write a simple Python function to calculate the factorial of a number using a for loop.	CO1
2	Write the first-order divided difference formula.	CO2
3	Differentiate between unconstrained and constrained optimization problems.	CO4
4	Define the primary goal of linear programming.	CO4
5	State the main idea behind genetic algorithms?	CO3
Section B (4 x 5 = 20 marks)		
6	Discuss the practical implications of choosing between numerical methods for root-finding based on the characteristics of the function and the problem at hand.	CO5

7	Given points (1, 1), (2, 4), and (3, 9), find the Lagrange interpolating polynomial.	CO2
8	Compare the advantages of using Lagrange multipliers over penalty function methods in handling constraints in optimization problems.	CO4
9	Discuss the key components of particle swarm optimization (PSO) and its application areas.	CO3
Section C (3 x 10 = 30 marks)		
10	Describe the bisection method for finding roots of a function. Write a Python function that implements the bisection method to find a root of the function $f(x) = x^2 - 4$ within the interval [1, 3].	CO5
11	(a) Explain Euler's method for solving ordinary differential equations. (b) Use Euler's method to approximate the solution of the differential equation $y' = x + y$, with initial condition $y(0) = 1$ and step size $h = 0.1$, up to $x = 0.3$.	CO3
Optional Question of Section C		
12	Compare and contrast simulated annealing and particle swarm optimization as metaheuristic algorithms. Discuss their principles, advantages, and disadvantages. Provide examples of optimization problems where each algorithm is effectively applied, and illustrate their working through pseudocode or flowcharts.	CO3
OR		
13	Analyze the impact of genetic algorithm parameters (population size, crossover rate, mutation rate) on the convergence and performance of the algorithm. Provide a comparative study with examples.	CO3